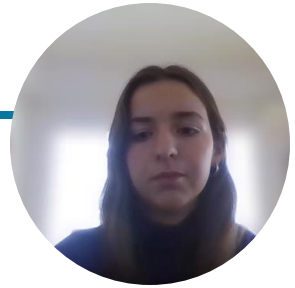


Youtube Video Link: https://youtu.be/WzzdKxyw6mA

Credit Application Governance Analysis

DEGO 2606 – Team 2

MSc Business Analytics | Nova SBE

Data Engineer
Sivert Ivarhus
70142@novasbe.pt

Data Scientist
Behnia Ghadiani
71819@novasbe.pt

Governance Officer
Pauline Kragset Pleym
74537@novasbe.pt

Product Lead
Laura Andrada
52013@novasbe.pt

Table of Contents

1 Introduction

2 Data Quality Findings

3 Bias Analysis

4 Privacy & Governance

5 Governance Recommendations

6 Conclusion

1 Introduction

The Scenario

NovaCred

A fintech startup using machine learning to automate credit decisions for loan applicants.

DataSet Overview

- 502 credit applications
- Nested JSON format

Our Mission

- Audit data quality
- Detect bias patterns in historical decisions
- Propose governance controls
- Demonstrate GDPR & EU AI Act compliance

Regulatory Context

GDPR	Data minimization, consent, right to erasure, storage limitation
EU AI Act	Credit scoring = HIGH RISK (Annex III) — conformity assessment required

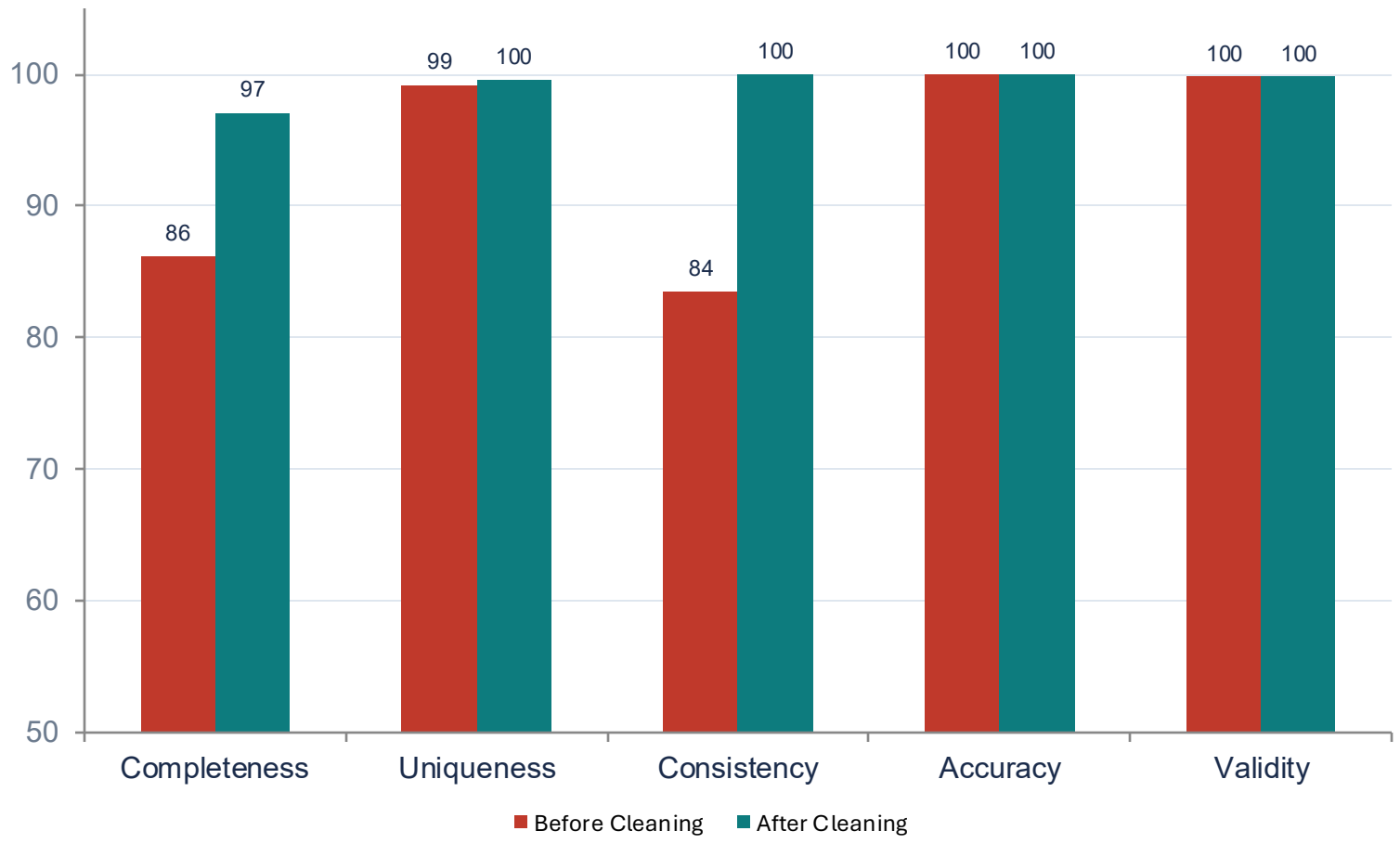
Dataset Schema

`applicant_info`	name · email · SSN · ip · gender · DOB · zip
`financials`	income · credit_months · DTI · savings
`spending_behavior`	category · amount (array of objects)
`decision`	approved · interest_rate · amount · reason

2 Data Quality Findings

Completeness	Uniqueness	Consistency	Accuracy	Validity
86.2% → 97.1%	99.2% → 99.6%	83.5% → 100%	100% → 100%	99.8% → 99.8%
4 cols >85% missing	4 duplicate app IDs detected	8 strings in annual_income converted to numeric	2 SSNs linked to multiple applicant names	1 negative savings value corrected
Missing tokens (NA, unknown) standardized	502 → 500 records after dedup	Mixed numeric types in financial variables	4 SSN records set to null to resolve conflicts	11 invalid email formats nullified
Structural missingness in decision fields	Each _id now uniquely identifies a loan	Gender & date formats standardized	Decision field logic validated — no issues	Financial variables within plausible range

2 Data Quality Scores - Before vs After Cleaning



Overall Average

Before

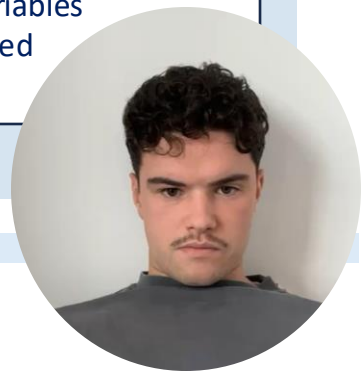
~~93.7%~~

After

99.3%

Outcome

500 records · 15 variables
PII fields removed

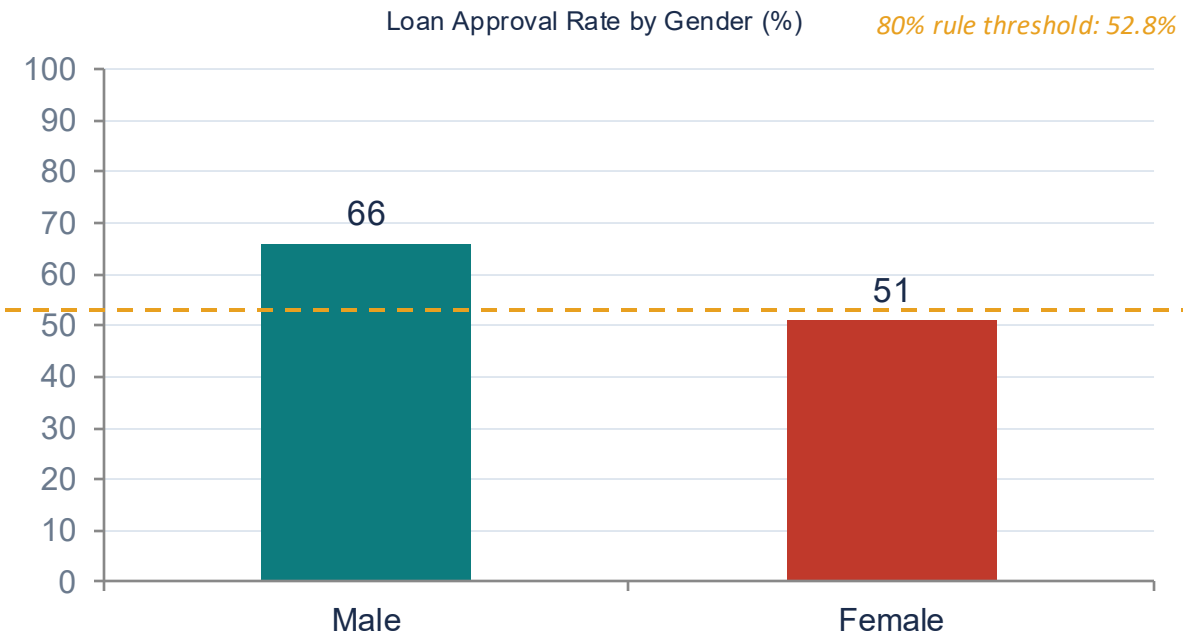


3 Bias Analysis - Gender

51%
Female Approval Rate
vs 66% for male

0.15
Dem. Parity Difference
z = −3.48, p = 0.0005

0.77
Disparate Impact Ratio
Below 0.8 threshold



Selection Bias: None

Gender is balanced: 50.4% female, 49.6% male ($\chi^2 = 0.03$, $p = 0.86$). No under/over-representation.

Outcome Bias: Significant

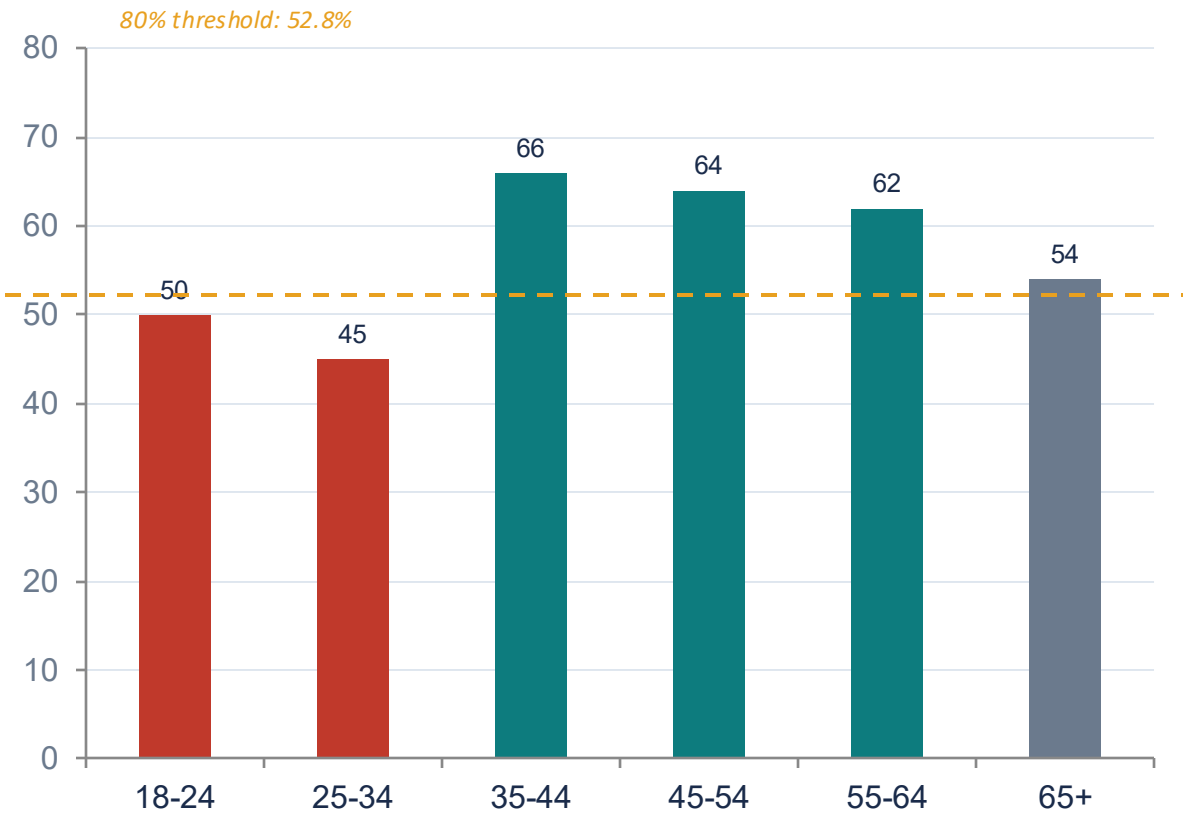
Female approval rate (51%) is below the 80% rule threshold (52.8%). DI ratio = 0.77 signals adverse impact.

Statistically Confirmed

Two-proportion z-test confirms the disparity is not random ($z = -3.48$, $p = 0.0005$). Systematic pattern detected.

3 Bias Analysis - Age, Proxy & Interactions

Approval Rate by Age Group



Age Bias Findings

- Selection bias, extreme groups are underrepresented ($\chi^2 = 296.46$, $p < 0.001$)
- Dem. Parity Difference across age groups = 0.22
- 80% rule violated for 18–24 (DI=0.75) and 25–34 (DI=0.67)
- Approval outcome differ significantly ($\chi^2 = 18.14$, $p = 0.003$ · Cramér's V = 0.19)

Proxy Discrimination

credit_history_months ↔ age
r = 0.652 ($p < 0.001$) · strong correlation with age, indicating a potential indirect proxy.

ZIP code ↔ gender
$p<0.001$, Cramér's V = 0.81 · statistical significance with strong association

Both credit_history_months and ZIP code have a significant impact on loan approval.
Significance proven through regression/chi-squared test

Age × Gender Interaction

No significant interaction found (all $p > 0.05$). Gender gap is consistent across all age groups - male approval always higher.

4 Privacy & Governance - EU AI Act & Controls

EU AI Act — Risk Classification

HIGH RISK

AI System

High-risk AI systems must implement risk management, data governance, logging, and human oversight under the EU AI Act.

Legal Reference:	EU AI Act — Annex III, Point 5(b)
Use Case:	Credit scoring / creditworthiness evaluation
Impact:	Automated decisions affect access to financial services
Requirement:	Conformity assessment required before deployment

Pseudonymisation Demo (SHA-256)

Original email:	`jerry.smith17@hotmail.com`
SHA-256 hash:	`116648a7761525746032d0ab...`

Original column removed — GDPR Art. 5(1)(c) data minimization applied

Governance Controls Proposed

01 Audit Logging
Log all automated credit decisions with timestamps, inputs, and outcomes. Supports GDPR Art. 5(2) & EU AI Act traceability.

02 Human Oversight
Rejected applicants must be able to request manual review. Required by GDPR Art. 22 and EU AI Act for high-risk systems.

03 Bias Monitoring
Conduct regular fairness assessments across gender & age groups. Prevent disproportionate impact on protected groups.

04 Data Retention Policy
Define and enforce retention schedules. Remove or anonymise PII once no longer needed — GDPR Art. 5(1)(e). Currently absent.

05 Transparency & Consent
Inform applicants of automated processing. Enable exercise of GDPR rights including the right to contest decisions.

4 Governance Recommendations

CRITICAL | ↗ Bias Analysis

1 Fix the Gender Bias Before Training

Re-weight or resample to balance approval outcomes

Remove or neutralise ZIP code — Cramér's V = 0.81 with gender

Target: DI ratio ≥ 0.80 before any model is trained

CRITICAL | ↗ Privacy Notebook

2 Implement Consent & Legal Basis Tracking

Log lawful basis (Art. 6) for every application processed

Add consent capture flow for automated decision-making

Currently absent — blocks GDPR compliance entirely

SHORT-TERM | ↗ Governance Controls

3 Deploy Audit Logging & Human Review Path

Log every automated decision: timestamp, inputs, outcome

Expose a manual review endpoint for rejected applicants

Required by GDPR Art. 22 and EU AI Act (high-risk obligation)

SHORT-TERM | ↗ Privacy Notebook

4 Enforce Data Retention & Minimization

Define retention schedule — 5-year maximum recommended

Schedule automated PII purge after retention window

Extend pseudonymisation to SSN and ip_address in prod

ONGOING | ↗ Bias Analysis

5 Run Quarterly Fairness Monitoring

Recalculate DI ratio and demographic parity each quarter

Alert if any group's DI drops below 0.80 threshold

Track age group 25-34 and 18-24 — currently most at risk

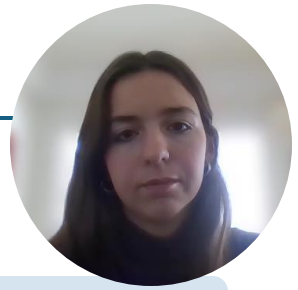
ONGOING | ↗ EU AI Act — Annex III

6 Conformity Assessment & EU AI Act Filing

Complete technical documentation before deployment

Register system in EU AI Act national database

Schedule annual re-assessment as data & model evolve



5 Conclusion

1 Data Quality

Overall quality improved from 93.7% → 99.3% through systematic cleaning across all 5 dimensions. Dataset reduced from 502 to 500 records; 21 to 15 columns.

2 Bias Detection

Gender DI = 0.77 → violates the 80% rule.
Younger applicants have significantly lower approval rates.
No age × gender interaction: the gender gap is consistent across age groups.

3 Privacy & GDPR

5 PII fields identified and removed or pseudonymised. Critical governance gaps remain: no consent mechanism, no data retention policy, no audit trail.

4 EU AI Act

NovaCred's credit-scoring ML system qualifies as HIGH RISK under Annex III. A conformity assessment, human oversight layer, and bias monitoring dashboard are mandatory.